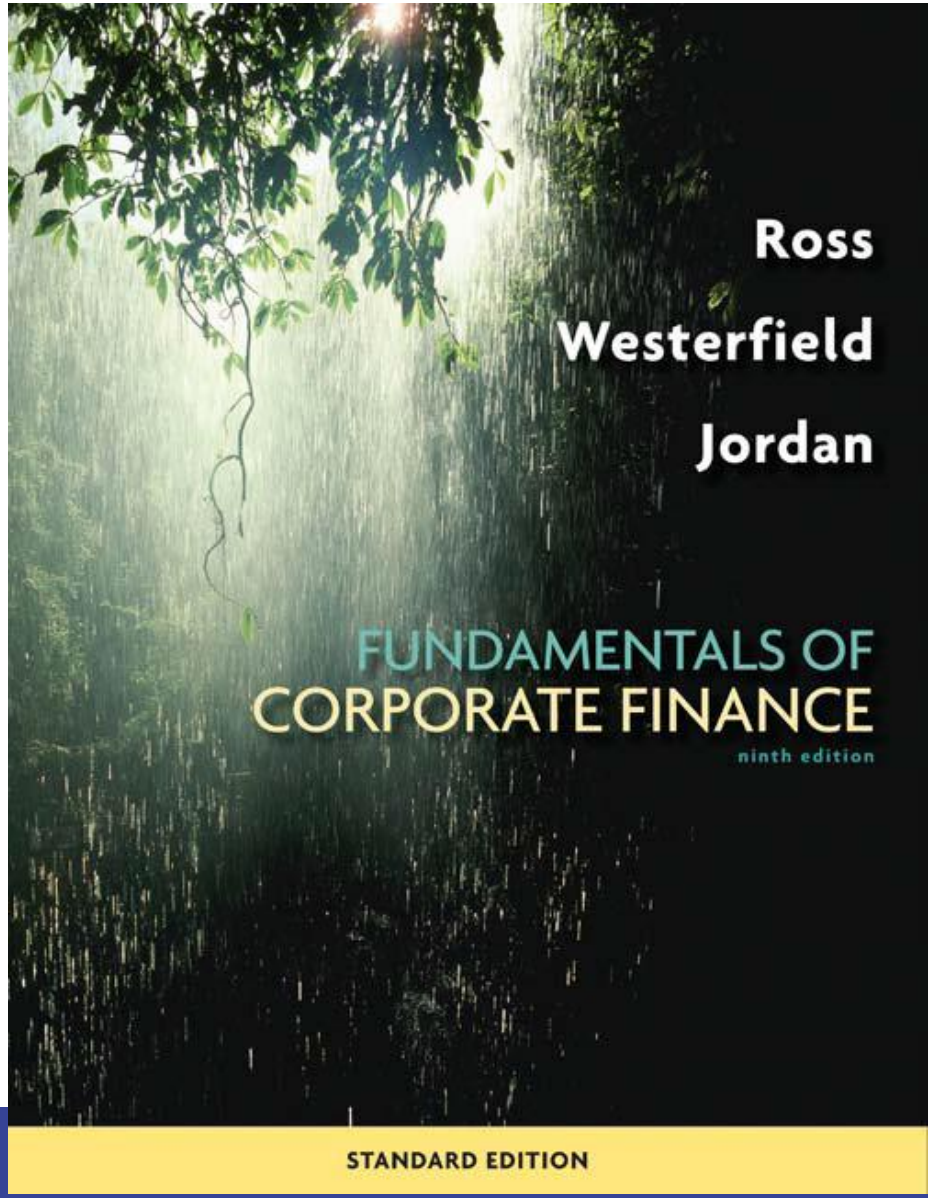
Advanced Corporate Finance

Lecturer:

Dr. Shaimaa Elgamal



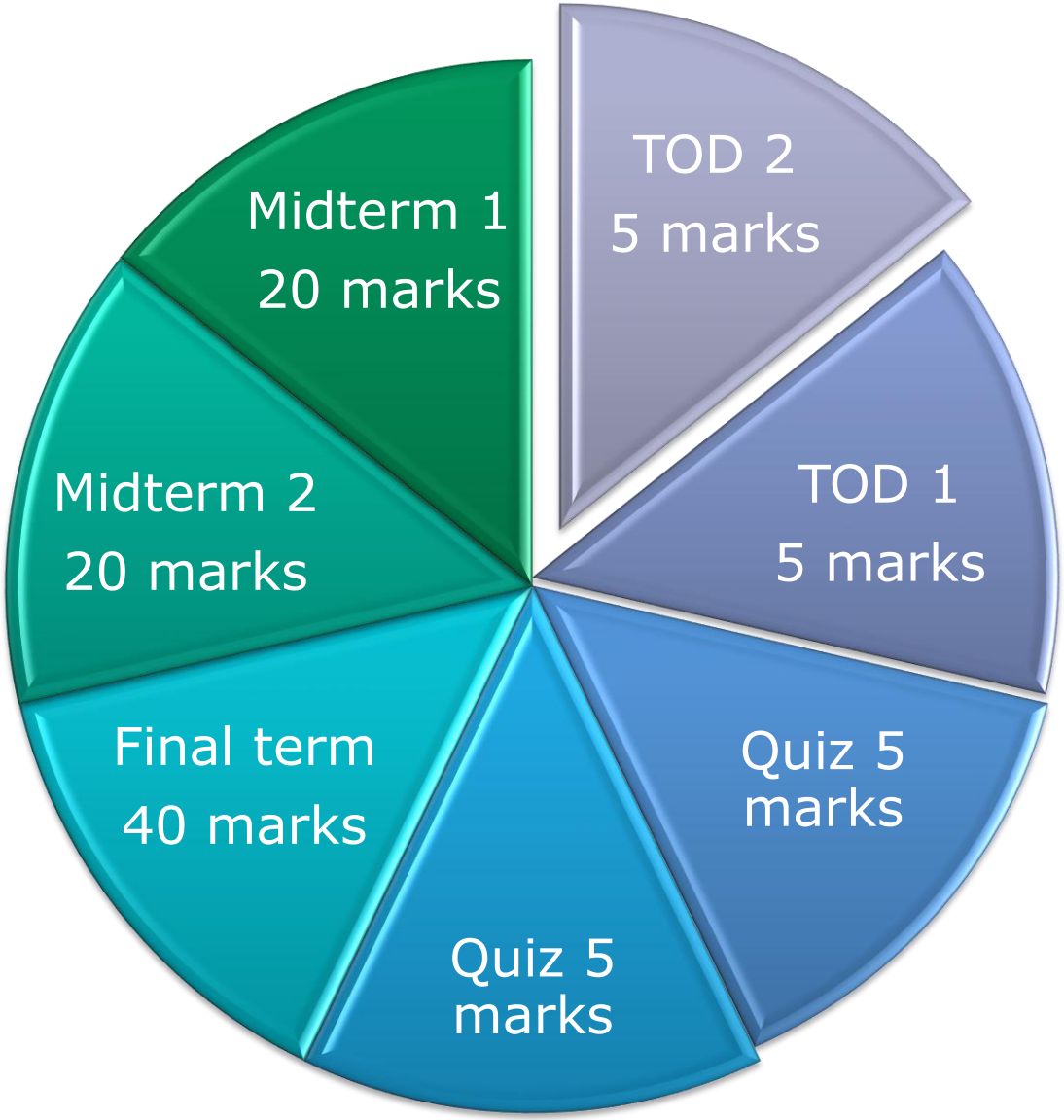
Class Policy & Discipline

Absence: After 4 times absence, student will be warned and repeated absences can affect his/her ability to fulfill the objectives of the module, and forced withdrawal will be implemented.

Extra curricula activities: students are in need to grant attendance to extra curricula activities, and bonus will be granted

Discipline: Students are in need to behave well at the class, and any violation to instructor command, will receive deduction of marks; and if repeated, student will be asked to leave the class and his/her attendance will be cancelled

Grades' Scheme



💰 You are the CFO

You work for an Egyptian company that has EGP 100 million available for investment.

The CEO tells you:

"You can choose ONLY ONE option. What should we do with the money?"

A. Expand the business
B. Buy another company
C. Invest in Egyptian stocks
D. Repay existing debt
E. Keep the money in the bank

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“Why did you choose that option?”

Because it gives higher returns

Because stocks are risky

Because debt is expensive

Because expansion increases profit

Because keeping cash is safer

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Breaking news: The Central Bank has increased interest rates significantly

"Would you change your decision?"

Breaking news 2: The Egyptian pound has depreciated significantly

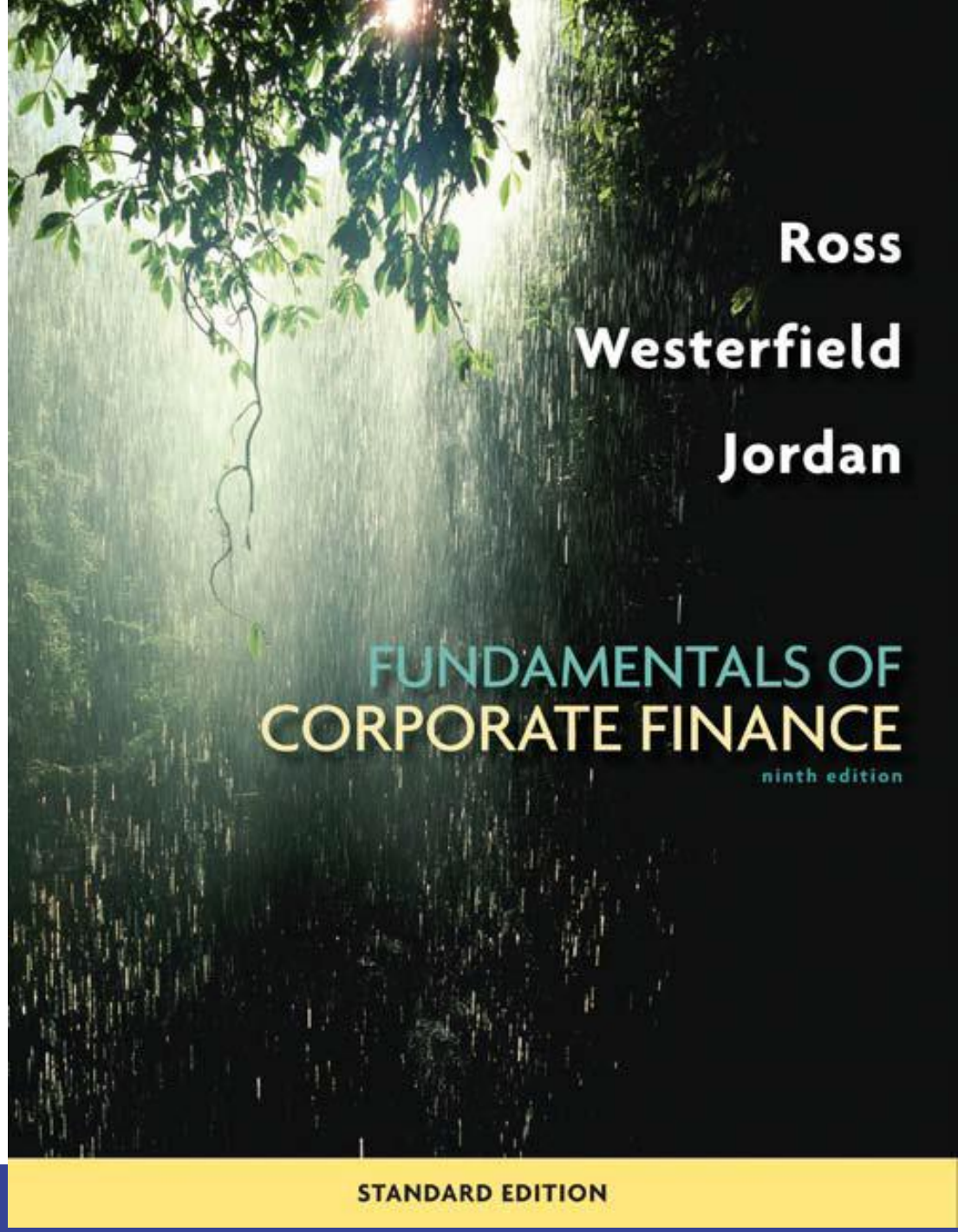
"Would you change your decision again?"

Breaking news 3: Inflation is increasing and the company's input costs are rising

"Would you change your decision again?"

Chapter 8:

Risk and Return Fundamentals



Learning objectives

1. Understand the meaning of risk, return, and risk preferences
2. Describe procedures for assessing and measuring the risk of a single asset
3. Discuss the measurement of return and standard deviation for a portfolio
4. Understand the risk and return characteristics of a portfolio
5. Explain the Capital Asset Pricing Model (CAPM) and the Security Market Line (SML)

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Formula list

- **Annual rate of return:** $r_t = \dfrac{(P_t - P_{t-1} + C_t)}{P_{t-1}}$
- **Range of returns:** Optimistic return−Pessimistic return
- **Expected return:** $\bar{r} = \sum_{j=1}^{n} r_j \times Pr_j$
- **Standard deviation of returns:** $\sigma_r = \sqrt{\sum_{j=1}^{n} (r_j - \bar{r})^2 \times Pr_j}$
- **Coefficient of variation:** $CV = \dfrac{\sigma_r}{\bar{r}}$

- **Portfolio return:** $r_p = \sum w_j r_j$
- **Portfolio weights:** $w_j = \dfrac{\text{Value of asset } j}{\text{Total portfolio value}}$
- **Portfolio beta:** $b_p = \sum w_j b_j$
- **Capital Asset Pricing Model (CAPM):** $r_j = R_F + [b_j \times (r_m - R_F)]$
- **SML slope:** rm−RF

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Risk and Return Fundamentals: Return Defined – An Example

Ramzi wishes to determine the return on two stocks that he owned during 2012: Apple Inc. and Wal-Mart.
At the beginning of the year, Apple stock traded for US$90.75 per share, and Wal-Mart was valued at US$55.33. During the year, Apple paid no dividends, but Wal-Mart shareholders received dividends of US$1.09 per share. At the end of the year, Apple stock was worth US$210.73 and Wal-Mart sold for US$52.84.

Required: calculate the annual rate of return (*r*) for each stock.

Apple: (US$0 + US$210.73 – US$90.75) ÷ US$90.75 = 132.2%

Wal-Mart: (US$1.09 + US$52.84 – US$55.33) ÷ US$55.33 = –2.5%

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Risk of a Single Asset– An Example

- Nadir Company, a custom golf equipment manufacturer, wants to choose the better of two investments, A and B. Each requires an initial outlay of US$10,000 and each has a *most likely* annual rate of return of 15 percent.

Investment A: Pessimistic 13% optimistic: 17%

Investment B: Pessimistic 7% optimistic: 23%

- Management recruited you to estimate the returns associated with each investment

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Asset A appears to be less risky than asset B. Because the range is the difference between the pessimistic and optimistic annual rate of return. Thus:

- Investment A range = 17%- 13% = 4%
- Investment B range= 23%- 7% = 16%
- The risk averse decision maker would prefer asset A over asset B, because A offers the same most likely return with a lower range (risk).

TABLE 5.2 Assets A and B

	Asset A	Asset B
Initial investment	US$10,000	US$10,000
Annual rate of return		
Pessimistic	13%	7%
Most likely	15%	15%
Optimistic	17%	23%
Range	4%	16%

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Exercise (1)

Dani Khalil, a financial analyst for Orange industries, wishes to estimate the rate of return for two similar risk investments, X & Y. dani's research indicates that the immediate past return will serve as reasonable estimates of future returns. A year later, investment X had a market value of 20,000$; investment Y has a market value of 55,000$. During the year, investment X generates cash flow 1,500$ and investment y generated cash flow of 6,800$. The current market values of two investments X & Y are 21,000 & 55,000 respectively.

A- calculate the expected rate of return on investments X & Y

B- Assuming that the two investments are equally risky, which one should Dani recommend? Why?

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Rate of return:

$$r_t = \frac{(P_t - P_{t-1} + C_t)}{P_{t-1}}$$

a. **Investment X** Return $= \dfrac{(\$21,000 - \$20,000 + \$1,500)}{\$20,000} = 12.50\%$

Investment Y: Return $= \dfrac{(\$55,000 - \$55,000 + \$6,800)}{\$55,000} = 12.36\%$

b. Investment X should be selected because it has a higher rate of return for the same level of risk.

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Risk Measurement for a Single Asset:

- **Expected value of a return (*r*)** is the average return that an investment is expected to produce over time.

$$\bar{r} = \sum_{j=1}^{n} r_j \times Pr_j \qquad (5.2)$$

r_j = return for the jth outcome

Pr_j = probability of occurrence of the jth outcome

n = number of outcomes considered

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Exercise (2)

Company X needs to choose between two assets (A &B), with probability of scenario analysis, each of them have the same probability at table 1

	Assets A & B
Optimistic	25%
Most likely	50%
Pessimistic	25%

The return for each asset, as following

	Asset A	Asset B
Optimistic	13%	7%
Most likely	15%	15%
Pessimistic	17%	23%

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Risk Measurement for a Single Asset: Expected Values of Returns

TABLE 5.3 Expected Values of Returns for Assets A and B

Possible outcomes	Probability (1)	Returns (2)	Weighted value [(1) × (2)] (3)
Asset A			
Pessimistic	0.25	13%	3.25%
Most likely	0.50	15	7.50
Optimistic	0.25	17	4.25
Total	1.00		Expected return 15.00%
Asset B			
Pessimistic	0.25	7%	1.75%
Most likely	0.50	15	7.50
Optimistic	0.25	23	5.75
Total	1.00		Expected return 15.00%

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Risk Measurement for a Single Asset: Standard Deviation of Returns

Standard deviation (σ_r) measures the dispersion around the *expected value.*

The expression for the standard deviation of returns, σ_r, is:

$$\sigma_r = \sqrt{\sum_{j=1}^{n} (r_j - \bar{r})^2 \times Pr_j} \qquad (5.3)$$

The higher the standard deviation, the greater the risk.

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Risk Measurement for a Single Asset: Standard Deviation of Returns

TABLE 5.4 The Calculation of the Standard Deviation of the Returns for Assets A and B[a]

j	r_j	$\bar{r}$	$r_j - \bar{r}$	$(r_j - \bar{r})^2$	Pr_j	$(r_j - \bar{r})^2 \times Pr_j$
Asset A						
1	13%	15%	−2%	4%	0.25	1%
2	15	15	0	0	0.50	0
3	17	15	2	4	0.25	1

$$\sum_{j-1}^{3}(r_j - \bar{r})^2 \times Pr_j = 2\%$$

$$\sigma_{r_A} = \sqrt{\sum_{j=1}^{3}(r_j - \bar{r})^2 \times Pr_j} = \sqrt{2\%} = 1.41\%$$

Asset B						
1	7%	15%	−8%	64%	0.25	16%
2	15	15	0	0	0.50	0
3	23	15	8	64	0.25	16

$$\sum_{j=1}^{3}(r_j - \bar{r})^2 \times Pr_j = 32\%$$

$$\sigma_{r_B} = \sqrt{\sum_{j=1}^{3}(r_j - \bar{r})^2 \times Pr_j} = \sqrt{32\%} = 5.66\%$$

Risk Measurement for a Single Asset: Coefficient of Variation

- The **coefficient of variation**, ***CV***, is a useful measure in comparing risks of assets with differing expected returns.

$$CV = \frac{\sigma_r}{\bar{r}} \qquad (5.4)$$

- A higher coefficient of variation means that an investment has more volatility (unpredictability) relative to its expected return.

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Risk Measurement for a Single Asset: Standard Deviation of Returns

TABLE 5.5 Historical Returns, Standard Deviations, and Coefficients of Variation for Selected U.S. Government Securities and Investments (1900–2009)

Investment	Average annual return	Standard deviation	Coefficient of variation[a]
Treasury bills	3.9%	4.7%	1.21
Treasury bonds	5.0%	10.2%	2.04
Common stocks	9.3%	20.4%	2.19

[a]Calculated by dividing the standard deviation in column 2 by the average annual return in column 1.

Source: Elroy Dimson, Paul Marsh, and Mike Staunton, *Triumph of the Optimists: 101 Years of Global Investment Returns*, Princeton, NJ: Princeton University Press, 2002.

Risk of a Portfolio

- In real-world situations, the risk of any single investment would not be viewed independently of other assets.
- New investments must be considered in light of their impact on the risk and return of an investor's portfolio of assets.
- The financial manager's goal is to create an **efficient portfolio,** a portfolio that maximum return for a given level of risk.

$$w_j = \frac{\text{Value of asset } j}{\text{Total portfolio value}}$$

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Problem

James owns:

- 100 shares of Wal-Mart at $55 = $5,500
- 100 shares of Cisco at $25 = $2,500
- Total portfolio = $8,000

Solution:

WMT=$5,500/$8,000=0.6875

CSCO=$2,500/$8,000=0.3125

If Wal-Mart returns 10% and Cisco returns 20%:

rp=0.6875(10%)+0.3125(20%)=6.875%+6.25%=13.125%

Portfolio Betas – An Example

$$b_p = (w_1 \times b_1) + (w_2 \times b_2) + \cdots + (w_n \times b_n) = \sum_{j=1}^{n} w_j \times b_j \qquad (5.7)$$

TABLE 5.10 Mazin Amin's Portfolios V and W

	Portfolio V		Portfolio W	
Asset	Proportion	Beta	Proportion	Beta
1	0.10	1.65	0.10	0.80
2	0.30	1.00	0.10	1.00
3	0.20	1.30	0.20	0.65
4	0.20	1.10	0.10	0.75
5	0.20	1.25	0.50	1.05
Totals	1.00		1.00	

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Portfolio Betas – An Example (cont'd)

The betas for the two portfolios, bv and bw, can be calculated as follows:

$$b_v = (0.10 \times 1.65) + (0.30 \times 1.00) + (0.20 \times 1.30) + (0.20 \times 1.10) + (0.20 \times 1.25)$$
$$= 0.165 + 0.300 + 0 .260 + 0.220 + 0.250 = 1.195 \approx \underline{1.20}$$

$$b_w = (0.10 \times .80) + (0.10 \times 1.00) + (0.20 \times .65) + (0.10 \times .75) + (0.50 \times 1.05)$$
$$= 0.080 + 0.100 + 0.130 + 0 .075 + 0.525 = \underline{0.91}$$

Conclusion:

Portfolio V (β = 1.20) is more risky than Portfolio W (β = 0.91)

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The CAPM: The Equation

Using the beta coefficient to measure nondiversifiable risk, the capital asset pricing model (CAPM) is given in the following equation:

$$r_j = R_F + [b_j \times (r_m - R_F)] \quad (5.8)$$

r_t = required return on asset j

R_F = risk-free rate of return, commonly measured by the return on a U.S. Treasury bill

b_j = beta coefficient or index of nondiversifiable risk for asset j

r_m = market return; return on the market portfolio of assets

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The CAPM: An Example

BenAli Corporation, a growing computer software developer, wishes to determine the required return on asset Z, which has a beta of 1.5. The risk-free rate of return is 7%; the return on the market portfolio of assets is 11%.

Substituting $b_Z = 1.5$, $R_F = 7\%$, and $r_m = 11\%$ into the CAPM yields a return of:

$$r_Z = 7\% + [1.5 \times (11\% - 7\%)] = 7\% + 6\% = \underline{13\%}$$

Interpretation: Investors require a 13% return on asset Z to compensate for its risk

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Example: CAPM with different BETAs

Asset	Beta	RF = 7%	rm = 11%	Required Return
A	0.5	7%	11%	7%+0.5(4%)=9%
B	1.0	7%	11%	7%+1.0(4%)=11%
C	1.5	7%	11%	7%+1.5(4%)=13%
D	2.0	7%	11%	7%+2.0(4%)=15%

Pattern: Higher beta = higher required return

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The Security Market Line (SML)

The SML is the graphical representation of the CAPM.

Key Points on the SML:

- **Y-intercept:** Risk-free rate (*RF*)
- **Slope:** Market risk premium (*rm*−*RF*)
- **X-axis:** Beta (nondiversifiable risk)

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Example- Plotting SML

Problem: Using *RF*=7% and *rm*=11%, plot the SML.

Beta	Required Return
0	7%
0.5	9%
1.0	11%
1.5	13%
2.0	15%

Shifts in the SML

1. Changes in Inflationary Expectations

If inflation increases, *RF* increases, shifting the SML upward in parallel

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Example: Inflation shift

Problem: Original: *RF*=7%, *rm*=11%, *bZ*=1.5, *rZ*=13%, and there is an expected increase in inflation by 3%

Solution:

New *RF*=10%, New *rm*=14%

rZ1=10%+[1.5×(14%−10%)]= 10%+6%=16%

Result: SML shifts up by 3% (parallel shift).

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Changes in risk aversion

If investors become more risk-averse, the market risk premium increases, causing the SML to steepen.

Example: Risk Aversion Shift

Problem: Original: RF=7%, rm=11%, bZ=1.5, rZ=13%. There is an expectation in market return increases to 14%.

Solution:

$rZ1=7\%+[1.5\times(14\%-7\%)]= 7\%+10.5\%=17.5\%$

Result: SML will steep

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Thank You….
See you next class